The web frontend part of microservice architectures and other distributed systems is often built as one or more single-page applications using frameworks like Angular, React, or Vue. Analyzing how SPAs are created is not in the scope of this book, but it’s essential to look at what changes are needed to support such frontend clients.

So far, you have interacted with the services composing the Polar Bookshop system through a Terminal window. In this section, we’ll add an Angular application that will be the frontend of the system. It will be served by an NGINX container and be accessible via the gateway provided by Edge Service. Supporting an SPA will require some additional configuration in Spring Security to address concerns like Cross-Origin Request Sharing (CORS) and Cross-Site Request Forgery (CSRF). This section shows how to do that.

11.5.1 Running an Angular application

The Polar Bookshop system will have an Angular application as the frontend. Since this book doesn’t cover frontend technologies and patterns, I have prepared one already. We just need to decide how to include it in the Polar Bookshop system.

One option is to let Edge Service serve the SPA static resources. Spring Boot applications serving a frontend usually host the source code in src/main/resources. That’s a convenient strategy when using template engines like Thymeleaf, but for SPAs like Angular, I prefer keeping the code in a separate module. SPAs have their own development, build, and release tools, so having a dedicated folder is cleaner and more maintainable. Then you can configure Spring Boot to process the SPA’s static resources at build time and include them in the final release.